

# Lecture 6 · Production II: Cost, Markups, Efficiency

MWG Ch. 5.C–5.D · cost functions, and the FOC that measures market  
power

Advanced Microeconomics 1

Thursday 15 October

## Why rebuild the theory around cost?

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Four reasons, in ascending order of importance:

1. Cost minimization is **necessary** for profit maximization — no extra assumption.
2. It is **well-posed** even under nondecreasing returns, where PMP explodes.
3. It is **where the data are**: firms report expenditures and input quantities, not production sets.
4. It remains valid **even when the firm has market power in its output market**: cost minimization only requires price-taking in *input* markets. This is the loophole through which the entire modern markup literature climbs — the centerpiece of today's second half.

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## **The cost minimization problem**

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## CMP (single output)

Given input prices  $w \gg 0$  and target output  $q$ :

$$c(w, q) = \min_{z \geq 0} w \cdot z \quad \text{s.t.} \quad f(z) \geq q.$$

Solution: **conditional factor demand**  $z(w, q)$ ; value: **cost function**  $c(w, q)$ .

- FOC (interior):  $w_\ell = \lambda f_\ell(z^*)$ , so  $\text{MRTS}_{\ell k} = w_\ell / w_k$  — isoquant tangent to isocost.
- The multiplier:  $\lambda = \partial c / \partial q =$  **marginal cost**. Keep this; it is the hinge of the markup derivation.
- Formally: this *is* the EMP with  $f$  playing  $u$  and  $w$  playing  $p$ . Every Ch. 3 result about  $(e, h)$  has a twin about  $(c, z)$  — next slide is copy-paste with new letters, and that is the point.

## Properties of $c$ and $z$

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### Proposition · MWG 5.C.2

For  $f$  continuous with nonempty upper contour sets:

1.  $c$  is **h.d.1** in  $w$  and **nondecreasing** in  $q$ ;  $z$  is **h.d.0** in  $w$ .
2.  $c$  is **concave** in  $w$ .
3. **Shephard's lemma**: if  $z(\bar{w}, q)$  is a singleton,  $\nabla_w c(\bar{w}, q) = z(\bar{w}, q)$ .
4. If  $z$  differentiable:  $D_w z = D_{ww}^2 c$  is **symmetric, NSD**, with  $D_w z w = 0$ .
5. If  $f$  h.d.1 (CRS):  $c$  and  $z$  are **h.d.1** in  $q$ ; if  $f$  concave,  $c$  is **convex** in  $q$  (nondecreasing marginal cost).

- Conditional demands slope down in own input price, symmetrically, *no compensation needed* — output is held fixed, which *is* the compensation.

## Trick #3 of 3: the duality trilogy, complete

*Concavity of  $c$  in  $w$ :* for fixed feasible  $\bar{z}$ ,  $w \mapsto w \cdot \bar{z}$  is linear;  $c(\cdot, q)$  is the lower envelope over feasible plans. Lower envelope of linear functions  $\Rightarrow$  concave.  $\square$

| Value function | Problem | Curvature in prices | Envelope: derivative         | Derivative matrix |
|----------------|---------|---------------------|------------------------------|-------------------|
| $e(p, u)$      | min     | concave             | $\nabla_p e = h$ (Shephard)  | $D_p h$ sym. NSD  |
| $\pi(p)$       | max     | convex              | $\nabla \pi = y$ (Hotelling) | $Dy$ sym. PSD     |
| $c(w, q)$      | min     | concave             | $\nabla_w c = z$ (Shephard)  | $D_w z$ sym. NSD  |

- One theorem, three costumes. If a prelim asks you to prove any row, you now prove all three.
- Empirical payoff of the third row: estimate a flexible  $c(w, q)$  (translog), differentiate, and you have the firm's entire input demand system — the workhorse of production econometrics since Christensen–Jorgenson–Lau.

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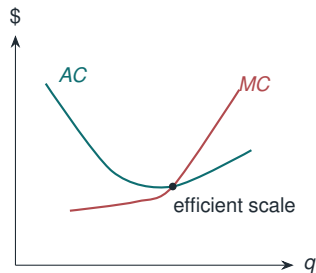
## **Geometry of cost and supply**

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## From $c(w, q)$ to the cost curves

Fix  $w$ ; write  $C(q) = c(w, q)$ ,  $AC = C(q)/q$ ,  $MC = C'(q)$ .

- $MC$  crosses  $AC$  at the minimum of  $AC$  (differentiate  $C/q$ ).
- **CRS:**  $c(w, q) = q \cdot c(w, 1) \Rightarrow AC = MC = \text{constant } \bar{c}(w)$ ; supply is flat at  $\bar{c}$ , quantity indeterminate — the GE-friendly case.
- Strictly convex  $C$ : competitive supply is  $p = MC(q)$  above the shutdown threshold ( $p \geq \min AVC$  short run;  $p \geq \min AC$  for long-run participation).
- Discipline: *fixed* vs. *sunk* is a timing statement, not a synonym pair — fixed costs you still avoid by exit; sunk costs you don't, and only the former belong in the shutdown rule.





## Short run vs. long run: LeChatelier

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- Short run: some inputs frozen at  $\bar{z}_2$ ;  $c^{SR}(w, q; \bar{z}_2) \geq c^{LR}(w, q)$ , with equality where  $\bar{z}_2$  is optimal — the long-run cost curve is the *lower envelope* of short-run curves. (Envelopes again.)
- **LeChatelier principle** (Samuelson): responses to price changes are weakly *larger in the long run*, when more margins can adjust. E.g. own-price input demand:  
$$\left| \partial z_\ell / \partial w_\ell \right|_{LR} \geq \left| \partial z_\ell / \partial w_\ell \right|_{SR}.$$
- Empirical echo: short-run vs. long-run elasticities differ systematically (energy demand, labor demand) — the theory says which way, for free.

## Worked example: Cobb–Douglas cost function

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$f(z) = z_1^\alpha z_2^\beta$ . Solve the CMP (do it once slowly; the answer is worth memorizing for  $\alpha + \beta = 1$ ):

$$c(w, q) = \gamma q^{\frac{1}{\alpha+\beta}} w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}}, \quad \gamma = (\alpha + \beta) \alpha^{-\frac{\alpha}{\alpha+\beta}} \beta^{-\frac{\beta}{\alpha+\beta}},$$

$$z_1(w, q) = \frac{\alpha}{\alpha + \beta} \cdot \frac{c(w, q)}{w_1}, \quad z_2(w, q) = \frac{\beta}{\alpha + \beta} \cdot \frac{c(w, q)}{w_2}.$$

- Checks: h.d.1 in  $w$ ; Shephard ( $\partial c / \partial w_1 = z_1$ ); cost shares constant at  $\alpha / (\alpha + \beta)$ ,  $\beta / (\alpha + \beta)$ .
- Returns to scale read off the exponent of  $q$ : marginal cost is increasing/constant/decreasing as  $\alpha + \beta \lessgtr 1$ .
- **Constant cost shares** are why Cobb–Douglas production functions and revenue-share arithmetic dominate the markup literature — next.

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**The markup: one FOC, one literature**

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## Deriving the markup from cost minimization

Let  $V$  be any **flexible** input (price  $P_V$ , quantity  $V$ ), chosen statically each period;  $Q = F(V, K, \dots)$ ; the firm may have *any* conduct in its output market. Cost minimization at output  $Q$ :

$$P_V = \lambda \frac{\partial F}{\partial V}, \quad \lambda = \text{marginal cost.}$$

Multiply both sides by  $V/(PQ)$  and define the **output elasticity**  $\theta_V = \frac{\partial F}{\partial V} \cdot \frac{V}{Q}$ :

The markup formula (Hall 1988; De Loecker–Warzynski 2012)

$$\mu \equiv \frac{P}{\lambda} = \theta_V \cdot \frac{PQ}{P_V V} = \frac{\theta_V}{s_V}, \quad s_V = \text{revenue share of input } V.$$

- Everything on the right is in the accounts — *except*  $\theta_V$ . The entire empirical fight is over one elasticity.
- Sanity checks: price taker  $\Rightarrow P = \lambda \Rightarrow s_V = \theta_V$ . Cobb–Douglas  $\Rightarrow \theta_V$  is a

## What the formula found

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### Application · De Loecker, Eeckhout & Unger (QJE 2020)

- Implementation on Compustat (US public firms, 1955–2016):  $V$  = cost of goods sold;  $\theta_V$  from estimated production functions by sector-year.
- Headline: the sales-weighted average markup rose from  $\approx 1.21$  in 1980 to  $\approx 1.61$  in 2016.
- Anatomy: driven by the *upper tail* (the median barely moves) and by *reallocation* of sales toward high-markup firms — not by every firm marking up more.
- Offered as a unifying suspect for macro puzzles: falling labor share, declining business dynamism, weak investment.
- Pause on the method: no demand estimation, no conduct assumption — just Ch. 5's FOC plus one elasticity. Maximal mileage from minimal theory. That is also its exposed flank.

## The fight over $\theta_V$

Bond, Hashemi, Kaplan & Zoch (JME 2021)

Compustat has *revenues*, not physical quantities. If the “production function” is estimated on revenue, you recover the **revenue elasticity**  $\theta_V^R$ , not  $\theta_V$  — and under imperfect competition  $\theta_V^R = \theta_V \cdot \frac{\varepsilon-1}{\varepsilon} = \theta_V/\mu$  (CES demand). Then

$$\hat{\mu} = \frac{\theta_V^R}{s_V} = \frac{\theta_V/\mu}{\theta_V/\mu} = 1 \quad \text{identically:}$$

a correctly-estimated revenue elasticity makes the markup estimator **uninformative by construction**. Whatever variation  $\hat{\mu}$  shows is coming from estimation error and specification, not market power.

- Further fronts: Traina (2018) — fold SG&A into variable cost and the *rise* largely flattens; input-market power contaminates  $P_V$ ; quantity-data studies (e.g. De Ridder–Grassi–Morzenti) find level and trend sensitive to revenue vs. quantity.

- The cleanest way to see all of this is a simulation where the truth is planted and the

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**Efficiency, misallocation, and what firms are for**

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## Efficient production

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### Definition · efficiency (MWG 5.F.1)

$y \in Y$  is **efficient** if there is no  $y' \in Y$  with  $y' \geq y$  and  $y' \neq y$ : no plan produces more of something with less of nothing.

### Proposition · MWG 5.F.1 — first-welfare flavor

If  $y \in y(p)$  for some  $p \gg 0$ , then  $y$  is efficient.

*Proof.* If  $y'$  dominated  $y$ , then  $p \cdot y' > p \cdot y$  (strictly positive prices), contradicting profit maximization.  $\square$

- Two lines, but load-bearing: *any* profit-maximizing plan at strictly positive prices wastes nothing — regardless of convexity, regardless of how peculiar  $Y$  is.



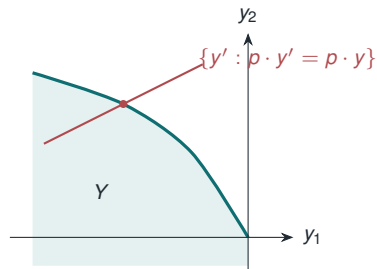
## The converse: supporting prices

### Proposition · MWG 5.F.2 — second-welfare flavor

If  $Y$  is **convex**, every efficient  $y \in Y$  is profit-maximizing for some  $p \geq 0, p \neq 0$ .

*Proof sketch.*  $y$  efficient  $\Rightarrow$  the open set  $\{y' : y' \gg y\}$  misses  $Y$ . Both convex  $\Rightarrow$  a **separating hyperplane** exists: some  $p \neq 0$  with  $p \cdot y \geq p \cdot \tilde{y}$  for all  $\tilde{y} \in Y$ ; the monotone side forces  $p \geq 0$ .  $\square$

- Caveats: some prices may be *zero*; and without convexity the theorem **fails** — efficient points inside a nonconvexity have no supporting price.



### Application · Hsieh & Klenow (QJE 2009)

- Flip Prop. 5.F.2 into an accounting device: if all firms faced the *same* prices, marginal revenue products would equalize. Firm-specific “wedges” (taxes, credit frictions, size-dependent policy) show up as **dispersion in TFP** — revenue productivity — within industries.
- Measure that dispersion in plant microdata; compute the TFP gain from reallocating inputs until marginal products equalize (US dispersion as the benchmark):  $\approx$  30–50% in Chinese manufacturing, 40–60% in India.
- “Misallocation”: the aggregate cost of *failing to find the supporting prices*. Care in reading: measurement error inflates dispersion (Bils–Klenow–Ruane 2021); treat magnitudes as upper-bound-flavored.

## What is the firm maximizing? (5.G)

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### Proposition · unanimity (informal)

If the firm is a price taker and owners' consumption possibilities depend on it only through their share of its *value*  $p \cdot y$ , then all owners — whatever their preferences — unanimously want profit maximized.

Where it breaks:

- **Market power:** owners who consume the output may prefer lower prices to higher dividends.
- **Incomplete markets:** profit is a random vector; without complete markets to price it, shareholders with different exposures rank plans differently — “maximize profit” is not even well-defined. (A hook for Ch. 6, next week.)
- Modern echo: shareholder-vs-stakeholder debates; Hart–Zingales (2017). The 1995 textbook aside is now a policy fight.

## Takeaways and readings

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### Takeaways

- CMP: valid under output-market power and nondecreasing returns;  $\lambda = MC$ ; the trilogy is complete:  $e, \pi, c$  — one envelope argument each.
- $\mu = \theta_V / s_V$ : one cost-min FOC turns accounting data into a market-power measure — and one elasticity carries all the identification risk (revenue vs. output).
- Profit max at  $p \gg 0 \Rightarrow$  efficient; the converse needs convexity and a separating hyperplane; the failure of common supporting prices is measurable (Hsieh–Klenow).

**Required:** MWG 5.C–5.D, 5.F–5.G.

**Applications:** De Loecker–Eeckhout–Unger (2020), Secs. I–III; Bond et al. (2021), Secs. 1–2; Hsieh–Klenow (2009), Secs. I–II.

**Monday: Exercise Class 2** (sheet issued today). Uncertainty begins Thursday week: skim MWG 6.A.